

Lab Assignment 4

Stacks

1. Develop a menu driven program demonstrating the following operations on a Stack using array:
(i) push(), (ii) pop(), (iii) isEmpty(), (iv) isFull(), (v) display(), and (vi) peek().

<https://www.javatpoint.com/ds-array-implementation-of-stack>

2. Given a string, reverse it using STACK. For example “DataStructure” should be output as “erutcurtSataD.”

<https://www.geeksforgeeks.org/stack-set-3-reverse-string-using-stack/>

3. Write a program that checks if an expression has balanced parentheses.

<https://www.geeksforgeeks.org/check-for-balanced-parentheses-in-an-expression/>

4. Write a program to convert an Infix expression into a Postfix expression.

<https://www.geeksforgeeks.org/stack-set-2-infix-to-postfix/>

5. Write a program for the evaluation of a Postfix expression.

<https://www.geeksforgeeks.org/stack-set-4-evaluation-postfix-expression/>

Additional Questions:

- Given an array A, find the nearest smaller element for every element A[i] in the array such that the element has an index smaller than i.

<https://www.interviewbit.com/problems/nearest-smaller-element/>

- Design a stack that supports getMin() in O(1) time and O(1) extra space.

<https://www.geeksforgeeks.org/design-a-stack-that-supports-getmin-in-o1-time-and-o1-extra-space/>

- <https://www.codechef.com/problems/BEX>